MILITARY STANDARD

[illegible] 23 [illegible]

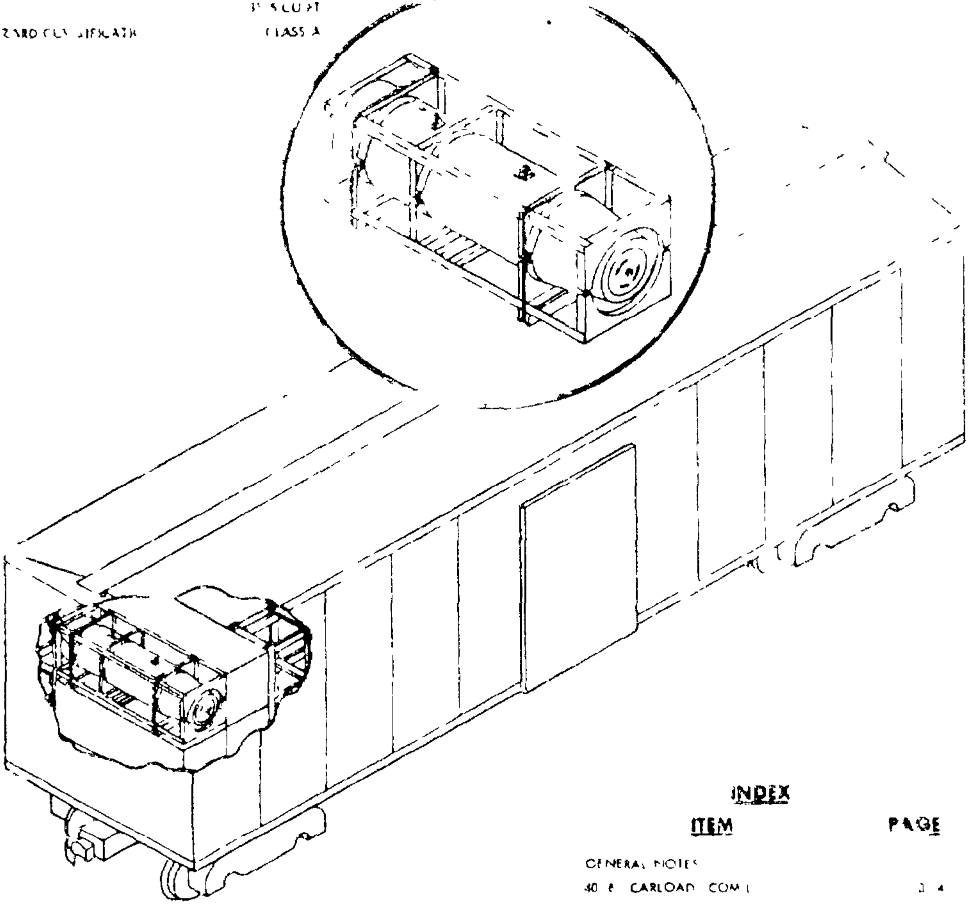
[illegible]

[illegible] UNIT [illegible] MK 23 AND [illegible]

[illegible] MK 23

UNIT LOAD DATA

LOAD DRAWING	WR 34/204
WEIGHT (APPROX)	SEE TABLE I
DIMENSIONS	33 1/4 L X [illegible]7/16 W X 31 1/4 H
CUBE	3[illegible] CU FT
HAZARD CLASSIFICATION	CLASS A



INDEX

ITEM	PAGE
GENERAL NOTES	
40 ft CARLOAD COMP	3 4
50 ft CARLOAD COMP	[illegible] B
50 ft CARLOAD COMP (ALTERNATE)	[illegible]

NOTES

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

2 FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE

W Gray NWHC 2/16/77

TECHNICAL DIRECTION AGENT (TDA) DATE

SYSCOM BY DIRECTION

APPROVED BY BUREAU OF EXPLOSIVES

11/[illegible]/77

SIGNATURE SUPERVISOR MILITARY AND [illegible] DATE

ORIGINATOR 12/6/77

DATE

NAVAL WEAPONS HANDLING [illegible]

PAGE OF

GENERAL NOTES

FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIAL, DESIGN AND INSTALLATION [illegible] SEE [illegible] REFERENCE DOCUMENT MIL-STD-1325 [illegible] RAILCAR LOADING OF HAZARDOUS MATERIALS

WHEN PLANNING SHIPMENTS [illegible] MINIMUM NUMBER OF CARS [illegible] CAPACITY REQUIRED FOR THE SHIPMENT. DODX UTILITY LOADER CARS SHALL NOT BE USED [illegible] EXPLOSIVES A OR B HAZARD [illegible]

LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES LENGTH, 112 INCHES INSIDE WIDTH

THE LOAD LIMIT OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE LOAD LIMIT IS CARRIED BY ONE SET OF TRUCKS

IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR

THE LOAD CONSISTS OF [illegible] MINE CASES MK 2[illegible] AND MOD [illegible] IN CRATE [illegible] 25 IN ACCORDANCE WITH WR 54 204

[illegible] UNIT LOADS ARE [illegible] AND LOADED WITH A SUITABLE FORK LIFT OR BOOM TRUCK

8 UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY)

ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 464? DATED 21 APRIL 197?

10 APPLICABLE MATERIAL SPECIFICATIONS

DUNNAGE LUMBER - FED SPEC MM-L-75
NAILS - FED SPEC FF [illegible] 10
STRAPPING - FED SPEC QQ-S-781 TYPE I HEAVY DUTY FINISH A B OR C
SEALS - FED SPEC QQ-S-781 STYLE II HEAVY DUTY

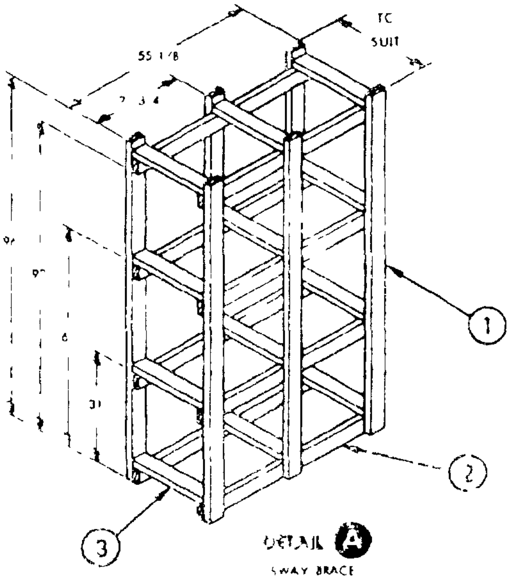
AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA. CLOSE AND SEAL BOXCAR DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I)

40 FT 6 IN BOXCAR, COMMERCIAL

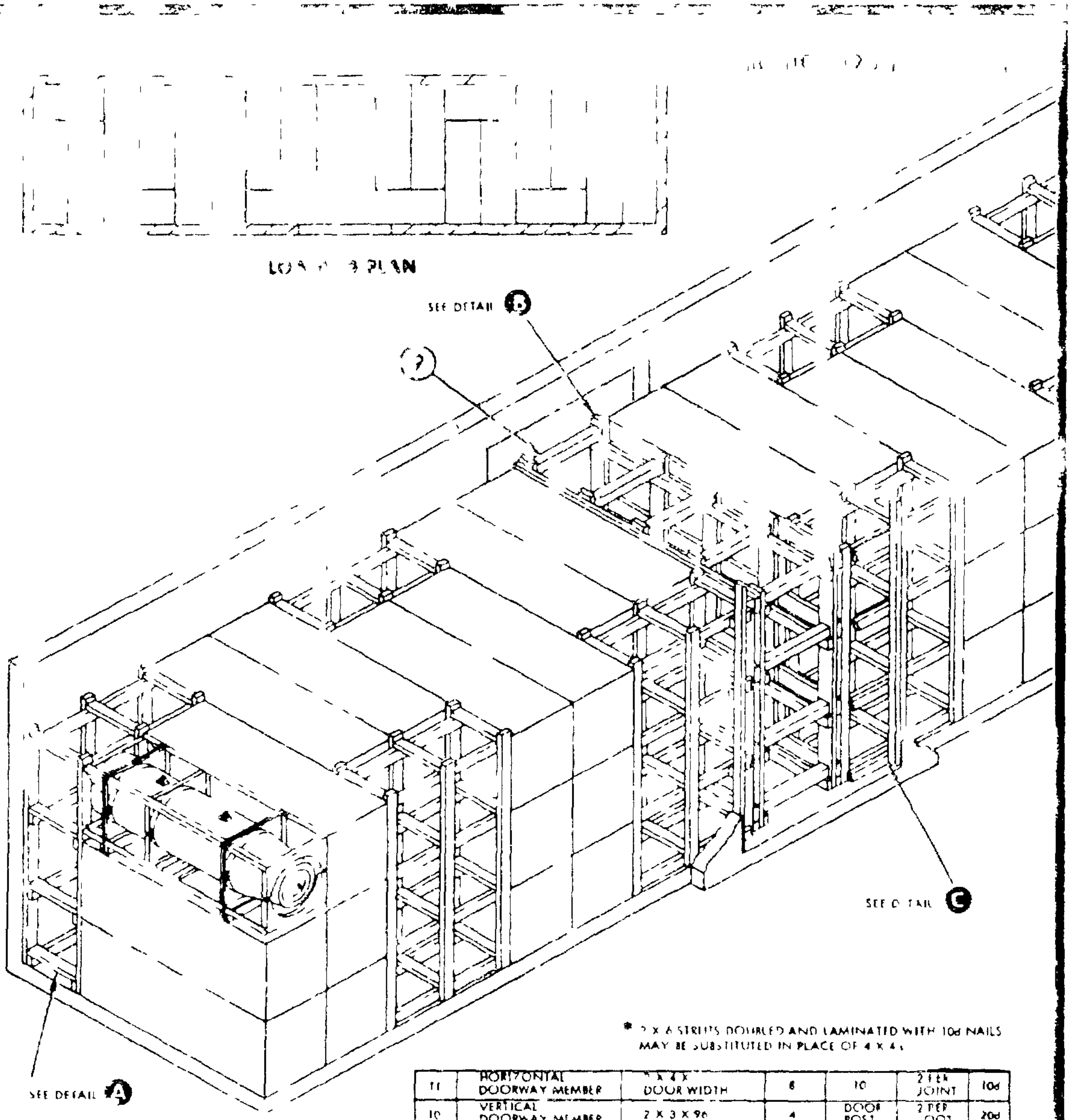
1 THE CARLOAD CONSISTS OF 48 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325 100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH. SEE MIL-STD-1325 (NAVY)

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE



DETAIL A
SWAY BRACE
(9 REQUIRED)



LOAD PLAN

* 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4s

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
11	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	8	10	2 PER JOINT	10d
10	VERTICAL DOORWAY MEMBER	2 X 3 X 96	4	DOOR POST	2 PER FOOT	20d
9	STRUT	4 X 4 X * WEDGE FIT	20	5	2 PER JOINT	16d
8	HOLD DOWN	X 3 X CUT TO SUIT	4	7	4	10d
7	CENTER GATE FILLER	X 3 X CAR WIDTH		[illegible]	2 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	X 4 X CAR WIDTH	8	[illegible]	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 X 6 X 96	10	SEE 4 & 6		
4	CENTER GATE HORIZONTAL	X 4 X CAR WIDTH	8		3 PER JOINT	10d
3	SWAY BRACE CROSS PIECE	2 X 4 X CUT TO SUIT	96	1	PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 55 B	64	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 96	48	SEE 2	-	

LIST OF MATERIALS & NAILING DATA

CARLOAD DATA

NUMBER OF UNIT LOADS 36

CARLOAD WEIGHT SEE TABLE I

TABLE I

[illegible]	UNIT LOAD WT (LBS APPROX)	CARLOAD WT (LBS APPROX)
[illegible]	2074	101,611
[illegible]	[illegible]	[illegible]

ESTIMATED WEIGHT OF [illegible] 12079 LB

INCLUDED IN CARLOAD WEIGHT

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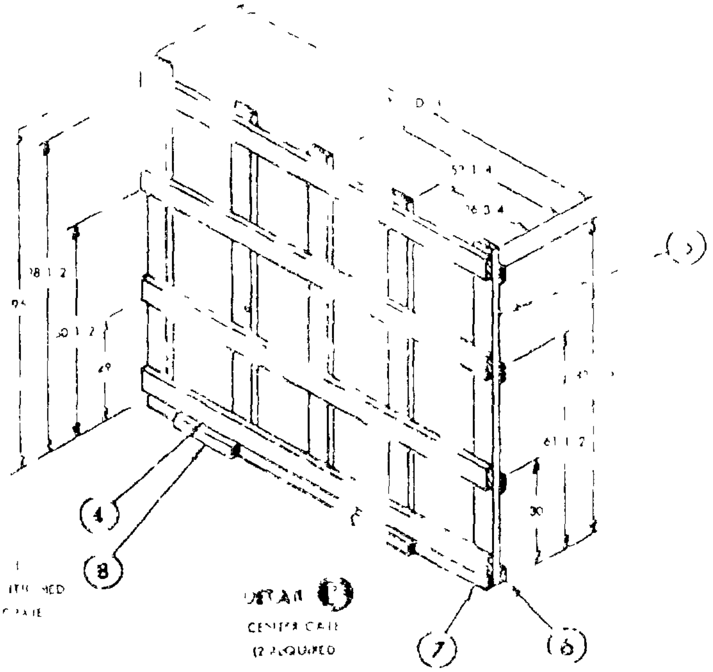
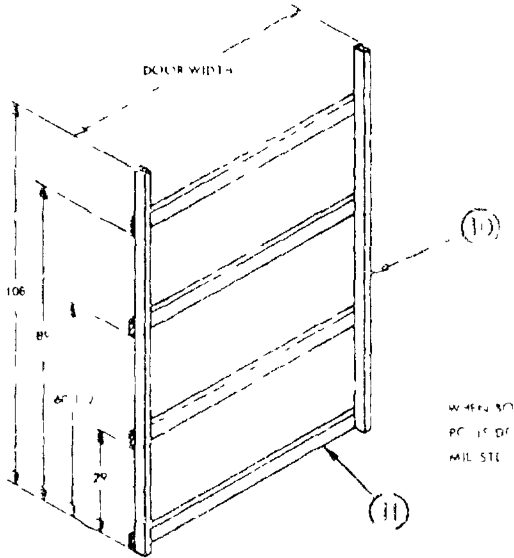

D
52 1/4
26 3/4
5
98 1/2
96
60 1/2
49
61 1/2
30
4
8
7
6

[illegible] ITCHED
[illegible] CRATE

DETAIL
CENTER GATE
(2 REQUIRED)


DOOR WIDTH
106
89
60 1/2
29
10
11

WHEN BOXCAR HAS STEEL DOOR
P[illegible] DOORWAY PROTECTION AS PER
MIL STD 132[illegible]

DETAIL 9
[illegible]
(2 REQUIRED)

[illegible]

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[illegible] DAY

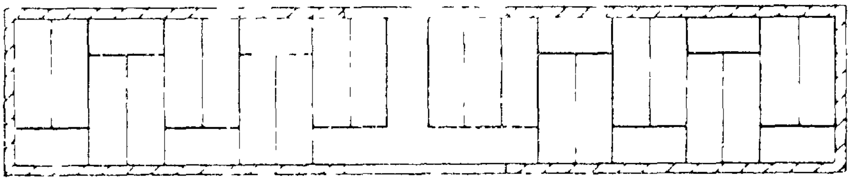
[illegible] OF 6 IN BOXCAR, COMMERCIAL

THE CARLOAD CONSISTS OF 67 UNIT LOADS WHICH MUST BE LOCATED [illegible] IN ACCORDANCE WITH THE PROCEDURAL DRAWING. BOXCARS [illegible] THAN [illegible] THE LOAD CARRYING CAPACITY [illegible] UNIT LOADS MAY BE LOADED WITH A LESSER AMOUNT. THE PROCEDURES SPECIFIED IN THIS DOCUMENT MUST BE FOLLOWED [illegible] POSSIBLE. IF A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED IN ACCORDANCE WITH [illegible] (NAVY) OR MIL-STD 1325-10? (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE [illegible]

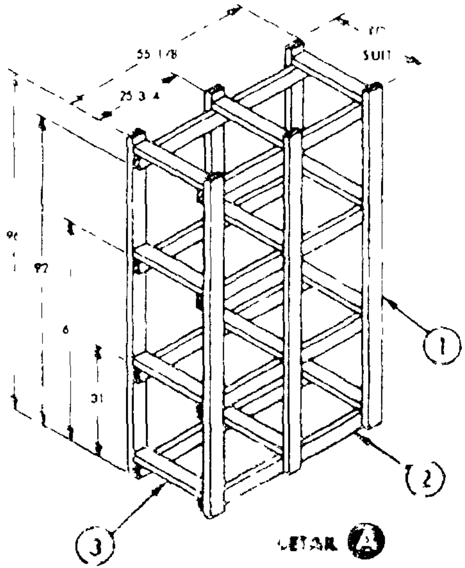
WHEN LESS THAN CARLOAD QUANTITIES ARE REQUIRED TO [illegible] COMMERCIAL BOXCAR AND A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF [illegible] BRACING AND OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-1G (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD 1325-1D (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH. SEE MIL-STD 1325 (NAVY).

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OT [illegible] THAN 10 FT WIDE.

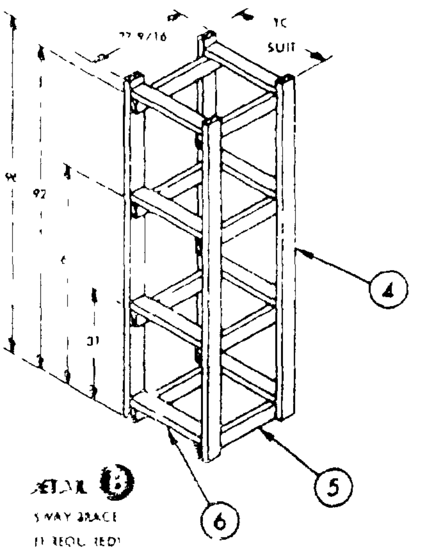
4 THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS PROVIDING [illegible] WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA HAVE TWO LATERALLY APPLIED 1 4 STEEL STRAPS PER STACK UNITIZING (ENCIRCLING) THE UNIT LOADS AND THE SWAY BRACE WITH EACH STRAP TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY [illegible] DOOR WHETHER MAIN OR AUXILIARY EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED [illegible] THE PLUG TYPE DOOR. DIMENSIONAL LUMBER DOORWAY PROTECTION PIECES 13 AND 14 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DIMENSIONAL LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN



DETAIL A
SWAY BRACE
(10 REQUIRED)

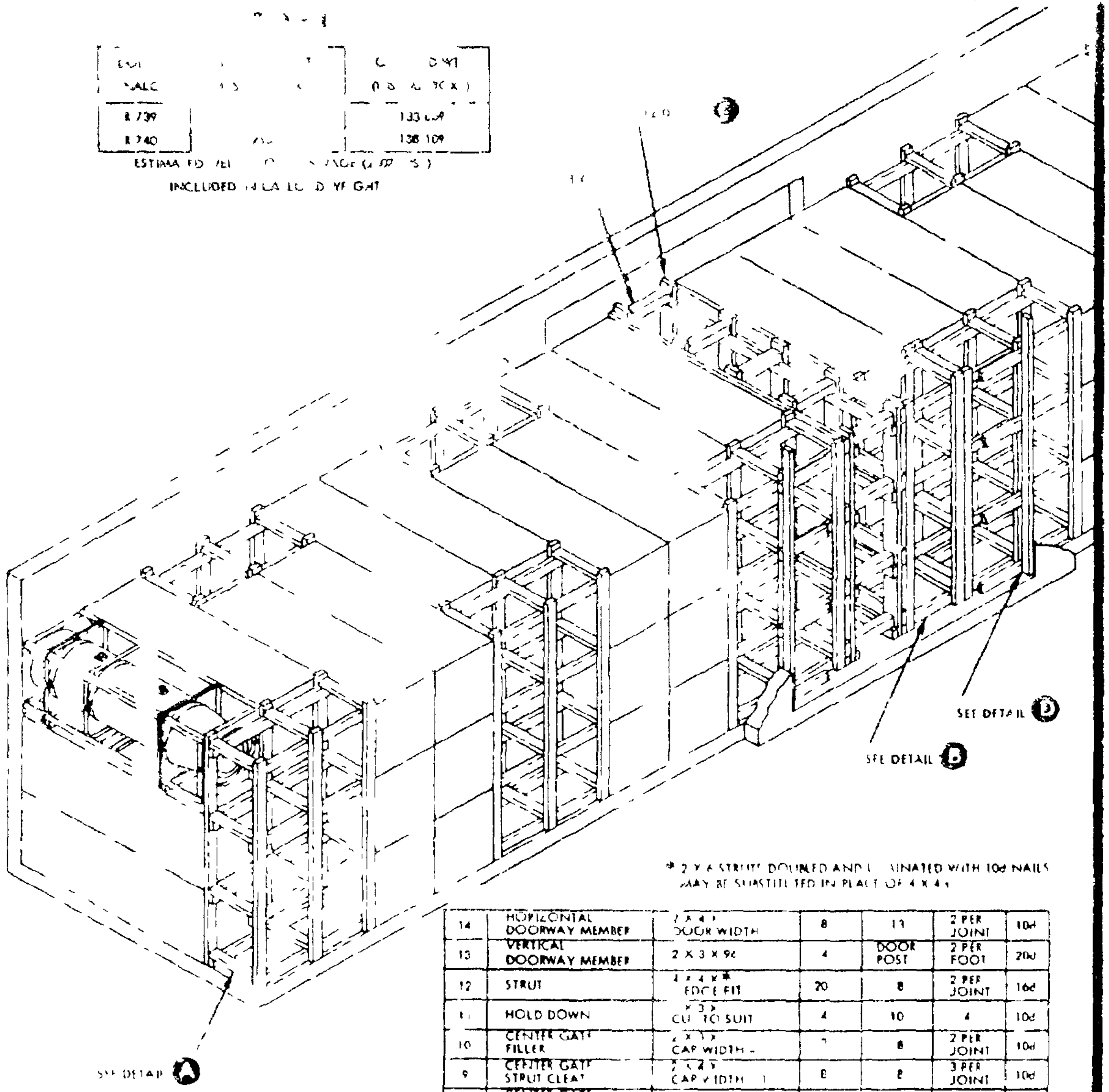


DETAIL B
SWAY BRACE
(1 REQUIRED)

[illegible]

CAR [illegible]	[illegible]	GROSS WT (LBS APPROX)
R 739		133 [illegible]
R 740	[illegible]	138 109

ESTIMATED WEIGHT OF DUNNAGE (LBS) INCLUDED IN CARLOAD WEIGHT



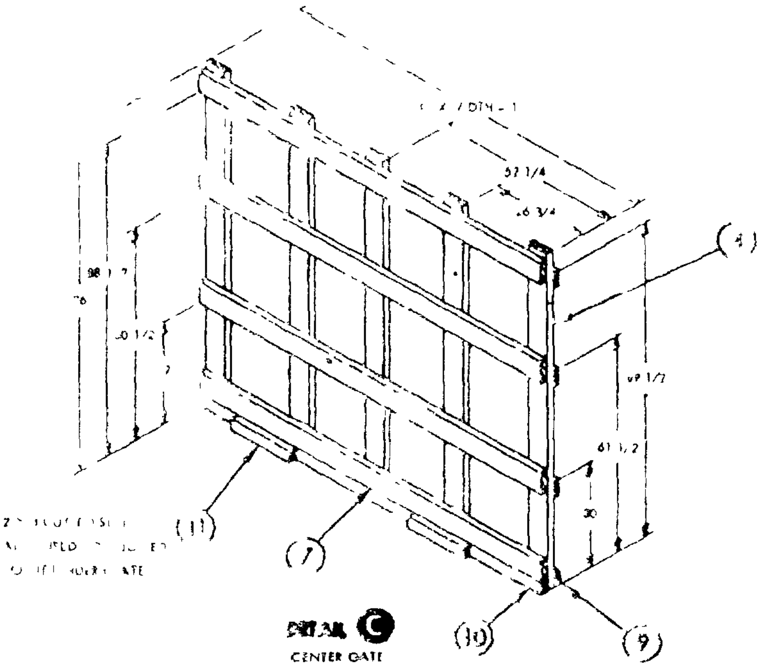
* 2 X 4 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4s

CARLOAD DATA

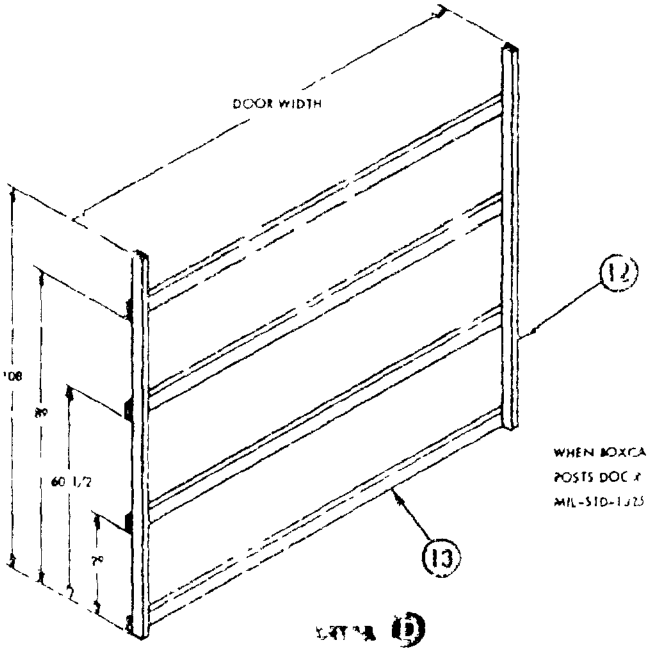
NUMBER OF UNIT LOADS 6[illegible]
CARLOAD WEIGHT SEE TABLE I

14	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	8	13	2 PER JOINT	10d
13	VERTICAL DOORWAY MEMBER	2 X 3 X 96	4	DOOR POST	2 PER FOOT	20d
12	STRUT	4 X 4 X* EDGE FIT	20	8	2 PER JOINT	16d
11	HOLD DOWN	2 X 3 X CUT TO SUIT	4	10	4	10d
10	CENTER GATE FILLER	2 X 3 X CAP WIDTH -	[illegible]	8	2 PER JOINT	10d
9	CENTER GATE STRUT CLEAT	2 X 4 X CAP WIDTH [illegible]	8	8	3 PER JOINT	10d
8	CENTER GATE VERTICAL	2 X 6 X 96	10	SEE 4 & 6		
7	CENTER GATE HORIZONTAL	2 X 6 X CAP WIDTH	8	8	3 PER JOINT	10d
6	SWAY BRACE CROSS BRACE	2 X 4 X CUT TO SUIT	8	4	2 PER JOINT	10d
5	SWAY BRACE HORIZONTAL	2 X 4 X 77 9/16	6	4	3 PER JOINT	10d
4	SWAY BRACE VERTICAL	2 X 4 X 96	4	SEE		
3	SWAY BRACE CROSS PIECE	2 X 4 X CUT TO SUIT	120	1	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 55 1/8	80	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 96	60	SEE		-
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE

LIST OF MATERIALS & NAILING DATA


57 1/4
26 3/4
98 1/2
76
30 1/2
69 1/2
61 1/2
30
4
11
7
10
9

DETAIL C
CENTER GATE
(2 REQUIRED)


DOOR WIDTH
108
89
60 1/2
29
12
13

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 [illegible] IS REQUIRED

DETAIL D
DOORWAY PROTECTION
(2 REQUIRED)

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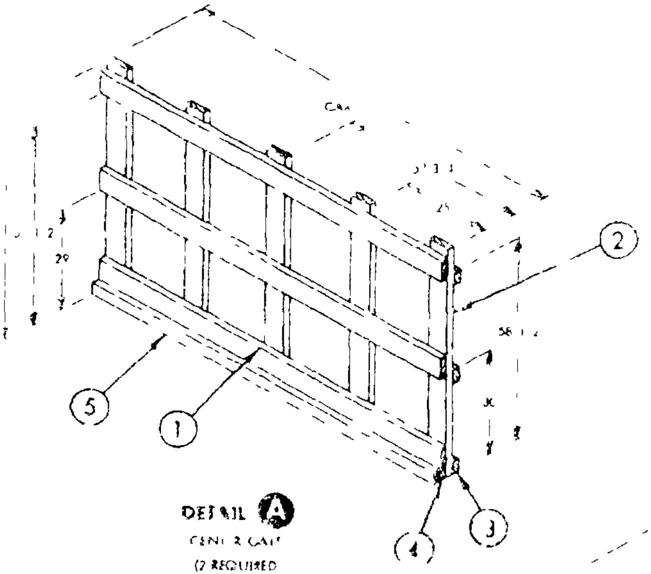
50 FT 6 IN BOXCAR, COMMERCIAL, ALTERNATE METHOD FOR 9 FT 4 IN WIDE BOXCAR

1 [illegible] OF 6 UNIT LOAD [illegible] IN ACCORDANCE WITH THE PROCEDURAL DRAWIN[illegible] REQUIRED [illegible] AND A PARTIAL LAYER RESULT [illegible] THE [illegible] CONSTRUCTED IN ACCORDANCE [illegible] SELECT THE TYPE OF [illegible] TO COMPLY [illegible] TO BE RETAINED [illegible] THE AAR MUST BE COMPLIED WITH [illegible] (NAVY).

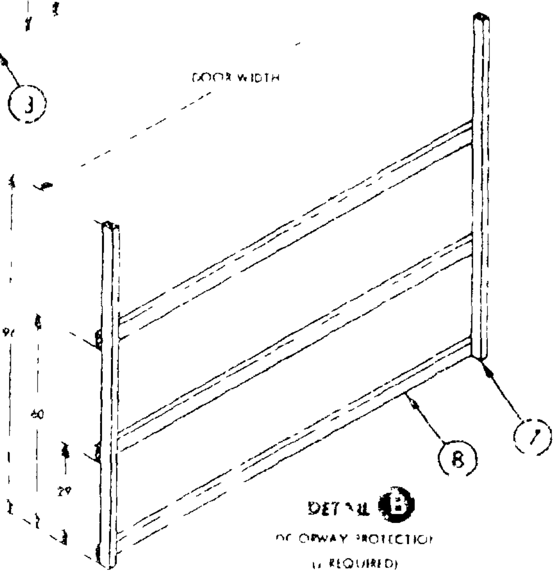
THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND [illegible] EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE [illegible] EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS [illegible] THAN 10 FT WIDE.

3 THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS PROVIDED STACKS WITH MORE THAN HALF OF THE UNIT [illegible] DOORWAY AREA [illegible] UNITIZED [illegible] LATERALLY APPLIED [illegible] 4 STEEL STRAPS PER STACK [illegible] TENSIONED AND [illegible] SEA[illegible] DOORWAY PROTECTION [illegible] DETAIL B IS NOT [illegible] WHEN [illegible] BOXCARS ARE USED EXCEPT [illegible] CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR [illegible] LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONA[illegible] DOOR MUST BE SECURELY [illegible] TOGETHER [illegible] A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY [illegible] THE WIRE ENDS TWISTED TOGETHER.



DETAIL A
CENTER GATE
(2 REQUIRED)

WHEN BOXCAR HAS STEEL DOOR
POSTS DOORWAY PROTECTION AS PER
MIL-STD-1325 (NAVY) IS REQUIRED



DETAIL B
DOORWAY PROTECTION
(2 REQUIRED)

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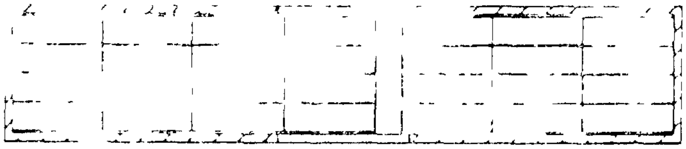
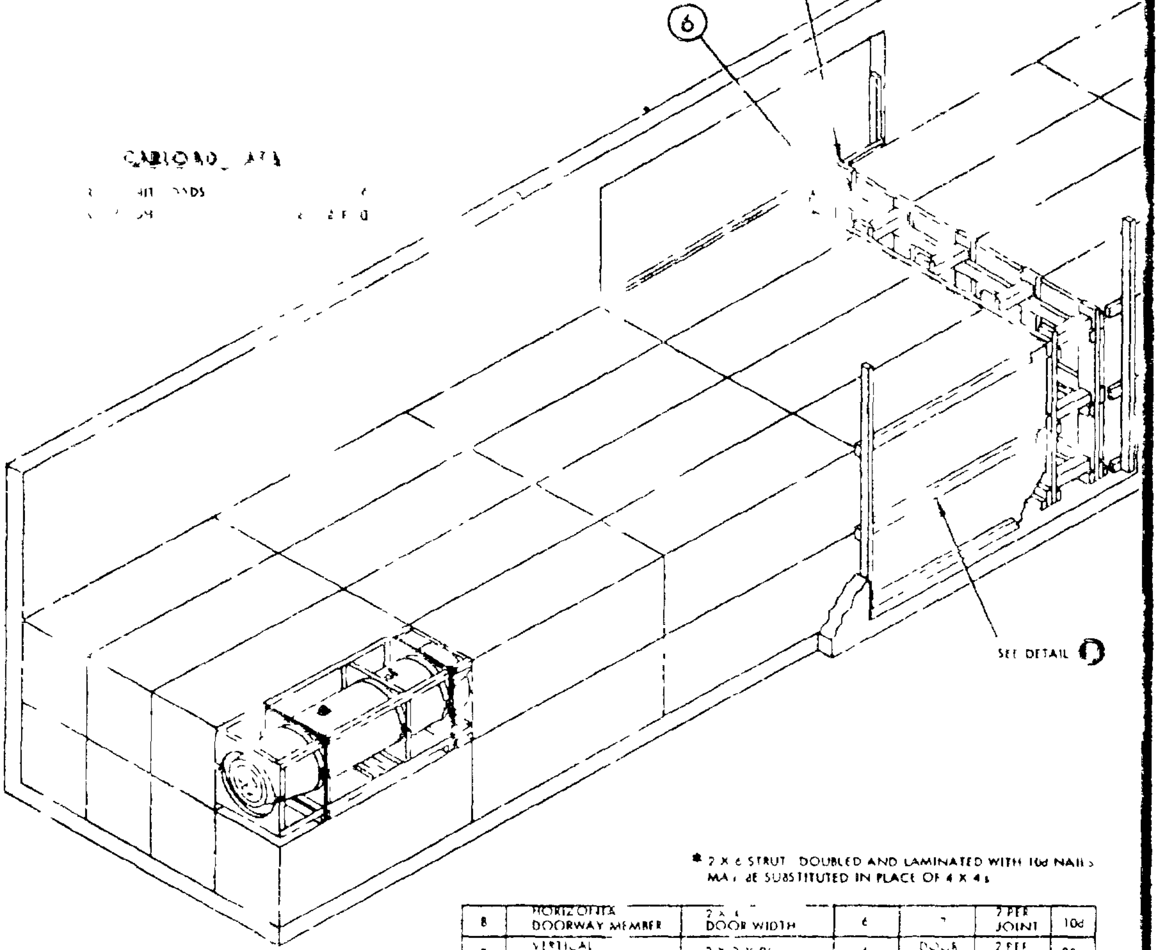


LOADING PLAN

CARLOAD DATA

[illegible] UNIT LOADS [illegible]

[illegible]



* 2 X 6 STRUT DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4s

PIECE NO	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
8	HORIZONTAL DOORWAY MEMBER	2 X 4 DOOR WIDTH	6	7	2 PER JOINT	10d
7	VERTICAL DOORWAY MEMBER	2 X 3 X 96	4	DOOR POST	2 PER FOOT	20d
6	STRUT	4 X 4 * [illegible]		2	2 PER JOINT	16d
5	HOLD DOWN	2 X 3 CAR WIDTH		4	1 PER FOOT	[illegible]
4	CENTER GATE FILLER	[illegible] CAR WIDTH			PER JOINT	10d
3	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH	[illegible]		3 PER JOINT	10d
2	CENTER GATE VERTICAL	2 X 6 X 66	10	SEE [illegible]		
1	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH	6	2	3 PER JOINT	10d

LIST OF MATERIALS & NAILING DATA

TABLE II

DODIC NALC	UNIT LOAD WT (LBS APPROX)	CARLOAD WT (LBS APPROX)
[illegible]739	2074	116,684
[illegible]740	2154	121,164

ESTIMATED WEIGHT OF DUNNAGE (540 [illegible]) INCLUDED IN CARLOAD WEIGHT

REVIEW ACTIVITY
NAVY OS

PREPARING ACTIVITY
NAVY OS
(PROJECT NO 8[illegible] - N[illegible])

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL ARS-1325-192 (NAVY)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802